CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Assignment 3: Supervised Machine Learning Algorithms

Course: Program Elective Course-I [23CS3507]
Category: Assignments | Assignment #3 | Academic Year: 2026-27

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1. Overview & Objective:

Mathematical derivation and algorithmic formulation of linear and non-linear classifiers.

2. Assignment Problems & Instructions:

1. Derive gradient descent update equations for Ordinary Least Squares OLS Linear Regression.
2. Derive the Logistic Regression sigmoid loss function binary cross-entropy .
3. Calculate Information Gain and Gini Impurity for Decision Tree node splitting.
4. Explain Support Vector Machines SVM maximal margin hyperplane and kernel trick.

Submission Guidelines & Verification:

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